HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained **Siemens TIA Portal** and **Step 7** PLC applications for automated control systems, supporting equipment integration and commissioning.
- Configured and integrated **Profinet**, **Modbus TCP**, and **Ethernet/IP** industrial communication protocols for seamless data exchange across automation hardware.
- Executed system testing, debugging, and performance optimization for automated equipment, ensuring reliable operation and adherence to project specifications.
- Provided on-site technical support and troubleshooting for installed automation systems at customer locations, resolving critical issues to minimize downtime.
- Contributed to software updates, conducted code reviews, and managed version control for PLC and HMI applications within an Agile development framework.
- Collaborated with electrical, mechanical, and software engineering colleagues to deliver end-to-end automation projects from design to deployment.
- Prepared comprehensive technical documentation, including test plans and system manuals, for new automation technologies and equipment.

SKILLS

PLC & HMI – Siemens TIA Portal, Step 7, Profinet, Modbus TCP, Ethernet/IP, IO-Link, SCADA

Programming Languages – Python, Java, C++, JavaScript

Cloud & IoT – AWS, Siemens Insights Hub, Industrial IoT

Engineering Concepts – Automation Control Systems, System Integration, Commissioning, Troubleshooting, Technical Documentation, Change Management

Software Development – Git, Docker, Agile Methodologies

PROJECTS

Siemens PLC Automation for Waste Treatment Equipment

- Developed PLC logic in **Siemens TIA Portal** for a new biohazardous waste treatment system, focusing on process control and safety interlocks.
- Integrated **Profinet** communication for sensor and actuator data, ensuring real-time feedback for system monitoring and HMI display.
- Assisted in the commissioning phase, performing initial tests and debugging to validate control system functionality against design requirements.

Python Scripting for Automation Control Deployment

- Created **Python** scripts to automate the deployment and configuration of control system software across multiple equipment units, reducing manual setup time by 40%.
- Implemented logging and error reporting within the scripts to track deployment status and identify configuration issues proactively.
- Developed reusable modules for common deployment tasks, enhancing the efficiency of the engineering team's rollout process.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)